

DBMS ASSIGNMENT

Online Food Delivery System

Database Design • Entities • Relationships

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Subject	Database Management Systems

A relational database design covering customers, restaurants, food items, orders, payments, delivery partners and addresses.

1. Name and Purpose of the Application

Application Name: Online Food Delivery System

Purpose:

An Online Food Delivery System is a database-based application that allows customers to browse restaurants, view food items, place orders, make payments, and track deliveries.

The database stores information about customers, restaurants, food items, orders, payments, delivery partners, and addresses, and connects these details through relationships.

2. List of Required Tables

1. Customer
2. Address
3. Restaurant
4. Food_Item
5. Category
6. Orders
7. Order_Item
8. Payment
9. Delivery_Partner
10. Delivery

3. Columns / Attributes for Each Table

3.1 Customer

Column	Description
Customer_ID	Unique ID of customer
Name	Customer name
Email	Customer email
Phone	Customer phone number
Password	Login password

Primary Key: Customer_ID

3.2 Address

Column	Description
Address_ID	Unique address ID
Customer_ID	ID of the customer
House_No	House/flat number

Street	Street name
City	City
Pincode	Postal code

Primary Key: Address_ID | Foreign Key: Customer_ID

3.3 Restaurant

Column	Description
Restaurant_ID	Unique restaurant ID
Restaurant_Name	Name of restaurant
Phone	Restaurant phone number
Location	Restaurant location
Rating	Restaurant rating

Primary Key: Restaurant_ID

3.4 Category

Column	Description
Category_ID	Unique category ID
Category_Name	Name of food category

Primary Key: Category_ID | Examples: Pizza, Burger, Indian, Chinese, Desserts

3.5 Food_Item

Column	Description
Food_ID	Unique food item ID
Restaurant_ID	Restaurant providing the food
Category_ID	Food category
Food_Name	Name of food
Price	Price of food
Availability	Whether food is available

Primary Key: Food_ID | Foreign Keys: Restaurant_ID, Category_ID

3.6 Orders

Column	Description
Order_ID	Unique order ID
Customer_ID	Customer who placed the order
Address_ID	Delivery address
Order_Date	Date of order
Total_Amount	Total order amount
Order_Status	Current status

Primary Key: Order_ID | Foreign Keys: Customer_ID, Address_ID

Examples of order status: Placed, Preparing, Out for Delivery, Delivered, Cancelled

3.7 Order_Item

Column	Description
Order_Item_ID	Unique order-item ID
Order_ID	Related order
Food_ID	Ordered food
Quantity	Quantity ordered
Subtotal	Price x quantity

Primary Key: Order_Item_ID | Foreign Keys: Order_ID, Food_ID

3.8 Payment

Column	Description
Payment_ID	Unique payment ID
Order_ID	Related order
Payment_Method	Method of payment
Payment_Date	Date of payment
Amount	Paid amount
Payment_Status	Payment status

Primary Key: Payment_ID | Foreign Key: Order_ID

Examples: UPI, Card, Cash on Delivery

3.9 Delivery_Partner

Column	Description
Partner_ID	Unique delivery partner ID

Partner_Name	Delivery partner name
Phone	Phone number
Vehicle_No	Vehicle number
Status	Availability status

Primary Key: Partner_ID

3.10 Delivery

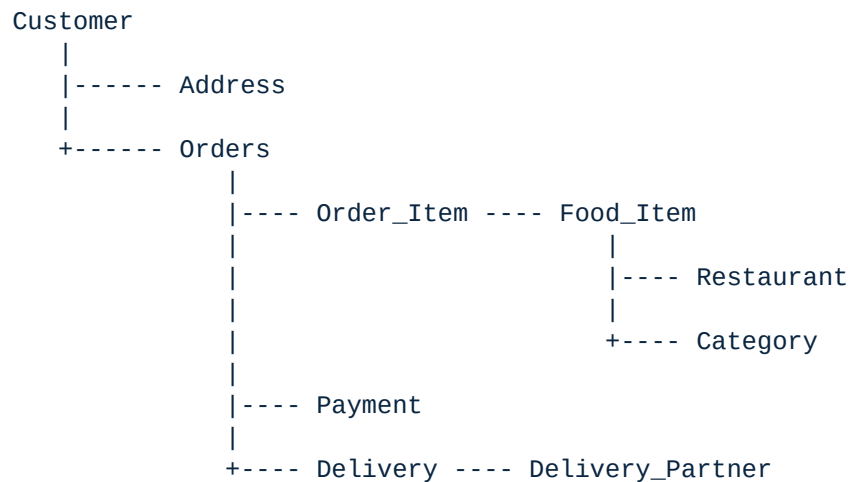
Column	Description
Delivery_ID	Unique delivery ID
Order_ID	Related order
Partner_ID	Assigned delivery partner
Pickup_Time	Food pickup time
Delivery_Time	Delivery completion time
Delivery_Status	Current delivery status

Primary Key: Delivery_ID | Foreign Keys: Order_ID, Partner_ID

4. Relationships Between Tables

- Customer → Address: One customer can have many addresses.
- Customer → Orders: One customer can place many orders.
- Restaurant → Food_Item: One restaurant can offer many food items.
- Category → Food_Item: One category can contain many food items.
- Orders → Order_Item: One order can contain multiple food items.
- Food_Item → Order_Item: One food item can appear in many orders.
- Orders → Payment: An order is associated with a payment.
- Orders → Delivery: An order can have delivery information.
- Delivery_Partner → Delivery: One delivery partner can deliver many orders.
- Address → Orders: One address can be used for multiple orders.

Simple Relationship Structure



5. Brief Explanation of Each Table

Table	Purpose
Customer	Stores customer account and contact information.
Address	Stores delivery addresses of customers.
Restaurant	Stores information about restaurants available on the platform.
Category	Stores different categories of food.
Food_Item	Stores food items, prices, availability, and their restaurant/category.
Orders	Stores details about orders placed by customers.
Order_Item	Stores individual food items included in each order.
Payment	Stores payment details for orders.

Delivery_Partner	Stores information about people who deliver orders.
Delivery	Tracks delivery details and delivery status.

Conclusion

The Online Food Delivery System demonstrates how a real-world application can be represented using a relational database. Each table stores a specific type of information, while Primary Keys and Foreign Keys connect the tables. This structure helps the system efficiently manage customers, restaurants, food items, orders, payments, and deliveries.